

PRIMORSKO-AKHTARSK, Russian Federation

WMO: 348240

Lat: 46.033N

Lon: 38.150E

Elev: 13

StdP: 14.69

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	6.7	12.4	-0.7	5.3	8.0	5.5	7.3	13.4	18.5	30.0	16.8	25.5	7.1	70	0.419

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.6	93.4	71.1	90.0	71.1	86.9	70.5	76.5	85.8	74.6	84.2	72.9	82.5	7.1	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
73.6	125.2	82.3	71.5	116.3	80.7	69.7	109.2	79.0	40.0	86.2	38.2	84.2	36.6	82.6	87.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.9	14.1	12.1	1.8	98.0	8.6	3.6	-4.3	100.6	-9.4	102.7	-14.2	104.8	-20.4	107.4		
(5)			1.0	79.6	8.4	2.9	-5.0	81.7	-9.9	83.4	-14.6	85.0	-20.7	87.0	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	54.0	30.7	32.7	41.3	53.0	64.8	72.3	77.5	76.0	66.0	54.0	42.6	35.0		
	DBStd	18.09	9.71	9.45	6.99	6.60	7.14	5.36	5.01	5.82	6.22	7.95	8.14	8.44		
	HDD50	2165	597	485	279	42	1	0	0	0	1	49	245	466		
	HDD65	5179	1062	903	734	362	95	8	0	1	62	349	672	931		
	CDD50	3607	0	2	10	133	458	670	853	804	480	174	22	1		
	CDD65	1147	0	0	0	3	88	228	388	341	90	9	0	0		
	CDH74	9290	0	0	0	22	583	1508	3637	3081	424	35	0	0		
	CDH80	3223	0	0	0	2	167	421	1399	1138	91	5	0	0		
	WSAvg	5.7	6.2	6.5	6.9	6.4	5.4	5.2	5.1	5.1	5.0	5.3	5.8	5.9		
	PrecAvg	22.2	2.0	1.5	1.4	1.5	1.9	1.9	2.6	2.0	1.7	1.5	1.9	2.5		
(15) Precipitation	PrecMax	29.0	5.7	3.5	3.0	3.7	5.7	5.7	10.6	6.9	7.2	4.4	4.6	4.4		
	PrecMin	15.8	0.2	0.2	0.2	0.0	0.4	0.2	0.2	0.4	0.0	0.1	0.2	0.4		
	PrecStd	4.0	1.5	0.9	0.8	0.9	1.3	1.3	2.2	1.7	1.7	0.9	1.1	1.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.2	57.9	63.9	77.0	88.9	92.1	97.9	99.0	88.0	78.7	66.4	55.5		
		MCWB	45.4	48.9	51.4	60.7	67.4	71.4	73.3	71.5	70.3	65.3	56.5	48.9		
	2%	DB	47.8	52.2	58.6	71.3	84.6	88.2	94.0	93.3	82.1	73.0	60.1	51.2		
		MCWB	43.7	46.2	48.9	57.5	66.3	70.7	71.4	70.2	66.1	62.0	54.2	47.4		
	5%	DB	44.3	47.7	54.7	67.2	80.4	84.8	90.7	89.4	78.6	68.4	56.4	48.0		
		MCWB	41.6	43.4	47.1	55.5	64.7	69.8	71.8	70.8	65.7	60.5	51.9	45.1		
	10%	DB	41.7	43.7	51.2	63.5	76.4	81.6	87.4	85.9	75.6	64.8	53.0	45.3		
		MCWB	39.7	40.7	45.3	54.0	64.2	68.6	71.3	70.4	64.6	58.2	49.6	43.0		
	0.4%	WB	46.8	50.3	52.4	62.1	71.1	76.7	80.4	78.2	73.9	67.5	57.8	50.4		
		MCDB	50.5	56.4	61.2	73.2	82.8	87.0	89.5	86.8	83.2	75.0	64.0	53.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	44.0	46.7	50.0	59.2	68.9	74.0	77.3	75.8	70.1	63.8	54.9	47.8		
		MCDB	47.1	51.2	57.3	68.6	79.6	83.4	86.6	85.1	77.7	70.4	59.1	50.8		
	5%	WB	41.9	43.7	47.8	56.9	67.0	71.9	75.2	74.1	68.2	61.3	52.4	45.3		
		MCDB	44.4	47.3	53.3	64.6	76.8	80.8	84.8	83.8	74.7	67.1	55.9	47.8		
	10%	WB	39.8	40.9	45.6	55.1	65.3	70.2	73.3	72.5	66.4	58.8	49.9	43.1		
		MCDB	41.7	43.5	50.6	62.0	74.3	78.8	82.8	82.4	72.7	63.5	52.7	45.2		
	5% DB	MDBR	8.9	10.6	11.5	13.6	14.1	12.9	14.6	15.2	14.6	12.7	10.4	8.7		
		MCDBR	11.7	15.9	18.0	20.1	20.4	18.3	20.0	20.0	19.1	18.1	16.0	12.9		
		MCWBR	8.8	11.1	10.4	9.7	9.3	8.3	8.5	9.4	9.8	10.2	10.5	9.8		
		MCWBR	11.2	15.0	16.5	16.2	17.2	14.5	15.9	15.7	16.1	16.2	15.4	12.3		
(35) Mean Daily Temperature Range	5% WB	MCWBR	9.0	11.0	10.3	9.0	8.9	8.1	9.3	9.5	9.7	10.2	10.6	9.9		
	Clear-Sky Solar Irradiance	taub	0.351	0.384	0.420	0.457	0.421	0.426	0.452	0.452	0.422	0.391	0.360	0.352		
		taud	2.274	2.198	2.142	2.085	2.295	2.321	2.253	2.237	2.288	2.323	2.337	2.278		
(42) All-Sky Solar Radiation	RadAvg	Ebn at Noon	228	243	252	254	269	268	259	253	250	239	225	213		
		Edh at Noon	26	34	42	48	40	40	42	41	36	30	25	24		
(44) All-Sky Solar Radiation	RadStd	RadAvg	343	625	983	1519	1941	2089	2078	1855	1354	830	433	284		
		RadStd	54	88	95	146	169	127	100	124	105	97	61	40		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (7 neighbors)		+1.12	N/A	N/A	N/A	N/A	N/A	N/A	-270	+225	+153	

Nomenclature: See separate page